

DEEPAK KUMAR

Birla Institute of Technology, Mesra, Ranchi

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SUMMARY

Backend-specialized Software Engineer with strong foundations in **Mathematics and Computing**. Proven track record of building scalable **MERN stack** applications and high-performance **Desktop Tools** using **Python/OpenCV**. Experienced in designing secure authentication systems (**JWT, OAuth 2.0**) and real-time communication architectures. Seeking to contribute research-driven solutions to innovative engineering teams.

EDUCATION

Birla Institute of Technology, Mesra

May 2027 (Expected)

Integrated M.Sc. in Mathematics and Computing | CGPA: 8.58/10

Ranchi, Jharkhand

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS.

TECHNICAL SKILLS

Languages: C++, Java, JavaScript (ES6+), Python, SQL,

Backend Database: Node.js, Express.js, REST APIs, JWT, MongoDB, MySQL

Frontend Libraries: React.js, Tailwind CSS, Cheerio, Redux

DevOps Tools: Git, GitHub Actions, Docker, Postman

EXPERIENCE

Full Stack Developer (Freelance)

Oct 2023 – Dec 2023

LooksLove Salon

Remote

- Delivered a production-ready booking system with secure **OTP-based login**, scaling to **500+ active users** and increasing digital appointment scheduling by **20%**.
- Built a comprehensive **Admin Dashboard** (React/Node.js) for real-time inventory tracking and staff management, reducing daily administrative overhead by 30%.

PROJECTS

LectureLens AI | *Python, OpenCV, NumPy, Tkinter*

[GitHub Link]

- Engineered a desktop video processing tool using **OpenCV** that removes **95%** of visual obstructions from lectures via a custom Progressive Background Building algorithm.
- Optimized rendering efficiency by **40%** using a **Dual-Resolution strategy** (processing low-res grayscale for motion analysis), generating 1080p PDF notes without quality loss.

Unlinked | *MERN Stack, Socket.io, JWT, Cloudinary*

[Live Demo]

- Architected a professional networking platform featuring **real-time messaging** and live notifications using **Socket.io**, supporting simultaneous interactions for **20+ users**.
- Implemented secure **JWT Authentication** and efficient **REST APIs** to manage complex social graph relationships (connections, likes) and optimized media storage via Cloudinary.

TnP Notifier | *Node.js, GitHub Actions, Cheerio, MongoDB*

[GitHub Link]

- Developed an automated placement alert system serving **200+ students**, eliminating manual monitoring and saving approximately **15 hours** of collective student time weekly.
- Orchestrated self-healing **GitHub Actions** CRON workflows with **99.9% uptime**, ensuring zero missed notifications for critical job announcements.

PUBLICATIONS

OAuth 2.0 Security Analysis | *Paper ID: 328 (Under Review)*

- Authored a research paper analyzing authorization code flow vulnerabilities; proposed a token integrity protocol to mitigate **CSRF** and **Replay Attacks**.

ACHIEVEMENTS

Problem Solving: Solved **400+ DSA questions** across various coding platforms (LeetCode, GFG).

IEEE Conference Presenter (2025): Presented poster on "OAuth 2.0 in Healthcare" at NIT Jalandhar.

[Certificate]

Startup Founder (2022-2023): Founded **Earthen Pots**, leading a team of 2 to manage production and D2C operations.

Volunteer (2022): International Mathematics Conference, assisting in event organization and logistics.

NCC 'C' Certificate: Demonstrated discipline, physical fitness, and leadership in annual training camps.